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METHOD STATEMENT FOR TESTING & COMMISSIONING OF LOCAL CONTROL PANELS (LCP) FOR ACMV EQUIPMENT

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1.0 Objective

To provide a systematic, step-by-step procedure for the inspection, testing, and commissioning of Local Control Panels (LCPs) for Air Handling Units (AHUs), Fresh Air Fans (FAFs), and Exhaust Air Fans (EAFs). The goal is to verify and document that the LCPs are installed safely, wired correctly as per approved drawings, and are fully functional in all operating modes (Hand, Off, Auto) before handing over for system integration.

2.0 Purpose

The Local Control Panel (LCP) serves as the dedicated "local brain" and primary electrical interface for a specific piece of ACMV equipment, such as an Air Handling Unit (AHU) or a Mechanical Ventilation Fan (FAF/EAF). It is a vital link that stands between the main building power supply, the mechanical motor it drives, and the centralized Building Management System (BMS).

3.0 Scope

This method statement covers all activities related to the testing and commissioning of ACMV LCPs, from initial physical inspection through to final functional testing. It is divided into two distinct phases:

- **Phase A:** Pre-Energization Checks ("Dead" Checks): All checks performed before the main power is supplied to the panel.
- **Phase B:** Energization & Functional Checks ("Live" Checks): All checks performed after the panel is safely energized.

4.0 References

- Project Practice for electrical installations.
- Project practice for air-conditioning and mechanical ventilation in buildings.
- Project Approved Shop Drawings: For electrical schematics, wiring diagrams, and panel layouts.
- Project Approved Material Submittals: For LCPs, VSDs, starters, and components.
- Manufacturer's Installation, Operation & Maintenance (IOM) Manuals.
- Project Health, Safety, and Environment (HSE) Plan & Risk Assessment.
- Workplace Safety and Health Act (WSH Act).

5.0 Responsibilities

- **Electrical Engineer/Supervisor:** To oversee all electrical work, ensure safety compliance, and authorize the final energization of the panel.
- **Commissioning Engineer:** To manage and coordinate the entire testing procedure, witness the tests, and verify the results against the design criteria.
- **ACMV/Electrical Technicians:** To execute the checks and tests as per this method statement under direct supervision.
- **Safety Personal:** To ensure all safety procedures, including Lock-Out/Tag-Out (LOTO), are strictly followed.
- **Controls (BMS) Contractor:** To be present during Phase B to assist with the verification of remote signals and interface points.

6.0 Equipment Required

- **Personal Protective Equipment (PPE):** Insulated gloves, safety glasses, safety footwear.
- **Lock-Out/Tag-Out (LOTO) Kit:** Including padlocks, hasps, and "Do Not Operate" tags.
- **Calibrated Test Equipment:**
 - Digital Multimeter
 - Insulation Resistance Tester ("Megger")
 - Clamp-on Ammeter
- **Hand Tools:** Insulated screwdriver set, torque wrench, wire strippers.
- **Documentation:** Approved copy of this Method Statement, relevant drawings, and commissioning checklists.



 Insulation Resistance Tester ("Megger")	Clamp Multimeter 	 Tools Box
 PPE	 Ladder	 Warning Signs

7.0 Step-by-Step Procedure

Phase A: Pre-Energization Checks ("Dead" Checks)

All work in this phase must be performed with the main power isolator to the LCP in the OFF position and secured with a Lock-Out/Tag-Out (LOTO) device.

Step 7.1: Safety Preparation & Documentation Verification

1. Obtain the necessary Permit-to-Work (PTW) if required.
2. The Electrical Engineer/Supervisor shall ensure the incoming power source is isolated, locked, and tagged.
3. Verify that the LCP model and all major internal components (breakers, VSD, contactors, overload relays) match the approved material submittals and drawings.

Step 7.2: Physical and Mechanical Inspection

1. Visually inspect the exterior and interior of the panel for any signs of physical damage sustained during transport or installation.
2. Check for cleanliness. Remove any construction debris, dust, or loose wire clippings from inside the panel.
3. Ensure the panel is securely mounted and level.
4. Verify that all internal components are firmly fixed to the backplate.

5. Check that all cable entries are properly glanded and sealed.
6. Confirm that all component labels and wire ferrules are present and match the details in the schematic drawings.

Step 7.3: Electrical Wiring and Termination Checks

1. Using a torque wrench or screwdriver, check the tightness of all power and control wiring terminations at breakers, terminal blocks, contactors, and VSDs. A loose connection is a critical fire hazard.
2. Verify that the incoming power cable sizes and the outgoing motor cable sizes are correct as per the drawings.
3. Visually inspect and confirm that the main earth wire is correctly terminated to the panel's earth bar and that the earth bar is bonded to the panel chassis.

Step 7.4: Insulation Resistance Test (Megger Test)

1. Isolate Sensitive Electronics: Disconnect the motor cables from the VSD/starter output terminals to test the motor and cable insulation separately from the panel's internal electronics.
2. Test Motor Cables: Using an insulation resistance tester set to 500V DC, test the insulation between:
 - Phase to Phase (L1-L2, L2-L3, L1-L3)
 - Phase to Earth (L1-E, L2-E, L3-E)
3. Record Readings: The resistance value should be very high, typically >100 Megaohms (MΩ). A low reading indicates a potential short circuit.
4. Reconnect Cables: Once the test is passed, securely reconnect the motor cables to the correct VSD/starter terminals.

Step 7.5: Control Circuit Continuity and Logic Checks

1. Using a multimeter in continuity mode ("beep test"), verify the logic of the control circuits:
 - **Emergency Stop:** Press the E-Stop button. Verify that the control circuit continuity is broken. Release the E-Stop and confirm continuity is restored.
 - **Switches:** Test the Hand-Off-Auto (H-O-A) selector switch to ensure it makes the correct contacts in each position as per the control schematic.
 - **Interlocks:** Verify any hardwired safety interlocks (e.g., fire alarm trip, filter differential pressure switch) are wired correctly to their designated terminals.

Phase B: Energization & Functional Checks ("Live" Checks)

Only authorized electrical personnel shall perform these checks. The panel door should remain closed whenever possible during live testing.

Step 7.6: First Energization

1. Ensure all personnel are clear of the equipment. Perform a final check inside the panel for any loose tools or materials. Close and secure the panel door.
2. The authorized Electrical Engineer shall remove the LOTO device and switch ON the main isolator supplying power to the LCP.
- 3. IMMEDIATELY use a multimeter to:**
 - Verify the incoming voltage (Phase-to-Phase and Phase-to-Neutral) is correct and balanced.
 - Confirm the control circuit transformer is providing the correct secondary voltage (e.g., 24V AC/DC).
 - Check for any "Phase Failure" or "Under/Over Voltage" alarms on the panel or VSD.
- 4. Record the voltage readings on the test sheet.**

Step 7.7: Local Control Functional Test ("Hand" Mode)

1. Turn the H-O-A selector switch on the LCP front panel to "HAND".
2. Press the local "START" button.
 - Verification: The fan/motor should start and run. The "RUN" indicator lamp should illuminate.
3. Using a clamp-on ammeter, measure the running current on all three phases at the motor terminals or VSD output. The readings should be balanced and not exceed the motor's Full Load Ampere (FLA) rating. Record these initial readings.
- 4. Press the local "STOP" button.**
 - Verification: The fan/motor should stop. The "RUN" lamp should extinguish.
5. While the motor is running in "HAND" mode, press the Emergency Stop button.
 - Verification: The motor must stop immediately. The LCP may show a "TRIP" or "FAULT" status.
6. Reset the E-Stop and any fault on the panel before proceeding.

Step 7.8: Remote Control Functional Test ("Auto" Mode)

1. Turn the H-O-A selector switch to "AUTO". The fan should remain OFF. This passes control to the remote system (typically the BMS).
2. Coordinate with the Controls Contractor to perform the following:
 - **Remote Start/Stop:** Send a "Start" command from the BMS. The fan should start. Send a "Stop" command. The fan should stop.
 - **Status Feedback:** Verify that the "Run Status" and "Fault/Trip Status" signals from the LCP are being received correctly by the BMS. Manually trip the overload relay to test the fault signal.
 - **VSD Speed Control (if applicable):** Send a speed control signal (e.g., 0-10V or 4-20mA) from the BMS. Vary the signal (e.g., 25%, 50%, 100%) and verify that the fan speed (and VSD's output frequency in Hz) changes accordingly.
 - Starting/stopping based on a time schedule.
 - Modulating fan speed via a 0-10V or 4-20mA signal to meet demand (e.g., based on static pressure or CO2 levels).
 - Shutting down during a building-wide fire alarm event.

8. Equipment Protection & Safety

This is one of the most critical functions of the LCP. It acts as a localized safety guard for the expensive motor and associated wiring.

- **Short Circuit Protection:** A Main Circuit Breaker (MCB or MCCB) provides instantaneous protection against dangerous high-current faults.
- **Overload Protection:** An overload relay continuously monitors the current drawn by the motor. If the motor is overworked and draws excessive current for a prolonged period (which would cause it to overheat and burn out), the overload relay will "trip," safely shutting down the motor.
- **Phase Protection:** For three-phase motors, the LCP often includes relays that protect against phase loss, phase sequence error, or phase imbalance, all of which can quickly destroy a motor.
- **Emergency Stop:** The LCP provides a prominent, hard-wired Emergency Stop button that will immediately shut down the equipment in case of a mechanical failure or safety incident, overriding all other commands.

9. Status Indication

The LCP provides immediate, at-a-glance visual feedback on the equipment's status via indicator lamps on the panel door:

- **RUN / ON Light (Typically Green):** Indicates that the motor is energized and running.
- **STOP / OFF Light (Typically Red):** Indicates that the motor is stopped.
- **TRIP / FAULT Light (Typically Amber/Yellow):** Indicates that the equipment has stopped due to a protective device tripping (e.g., motor overload, phase failure), alerting technicians to a problem.

10. Interface & Integration

The LCP acts as the bridge or gateway between the high-power electrical world and the low-voltage control world of the BMS.

- **Receives Commands:** It receives low-voltage signals from the BMS (e.g., a "start" command or a speed control signal) and translates them into high-power actions (e.g., closing a contactor to start the motor or telling the VSD to run at 35 Hz).
- **Provides Feedback:** It provides crucial feedback signals back to the BMS, such as:
 - **Run Status:** A "volt-free" contact that closes when the motor is proven to be running, confirming the command was executed.
 - **Fault Status:** A signal that alerts the BMS that the unit has tripped on a fault, allowing for a remote alarm to be generated.

11. Documentation and Handover

1. Throughout the process, record all readings, observations, and test results on the official Commissioning Test Sheet for the LCP.
2. Any failures, discrepancies, or non-conformities shall be recorded on a "Punch List" for rectification.
3. Upon successful completion of all tests, the Commissioning Engineer and Electrical Supervisor will sign the test sheet, confirming the LCP is safe and fully functional.
4. The completed and signed documentation will be included in the final Testing & Commissioning report for handover.

12. Health and Safety

- **Risk Assessment:** A comprehensive risk assessment must be conducted and understood by all team members before any work commences. This assessment will identify potential hazards and establish robust control measures.
- **Permit-to-Work (PTW):** A PTW for electrical works must be obtained from the main contractor or client's authorized representative before starting any "live" testing activities.
- **Authorized Personnel:** Only trained, qualified, and authorized electrical technicians and engineers are permitted to open, inspect, or test LCPs.
- **Toolbox Talk:** A daily toolbox talk or pre-task briefing will be held to discuss the specific scope of work, potential hazards for the day, and the required safety procedures.
- **Hazard: Electric Shock**
 - Control:** Never work on live circuits unless absolutely necessary (e.g., for voltage and current measurements). Use properly insulated tools. Never defeat safety interlocks. Always assume a circuit is live until proven otherwise with a multimeter
- **Application:** LOTO must be applied to the main upstream isolator feeding the LCP **BEFORE** any "dead" checks or physical work inside the panel is performed.
- **Clear Workspace:** The area in front of the LCP must be kept clear and unobstructed at all times to ensure quick egress in an emergency.
- **First Aid:** All team members must know the location of the nearest first aid kit and who the designated first-aiders are on site.
- **Fire Extinguisher:** A CO2 or Dry Powder type fire extinguisher suitable for electrical fires must be located near the work area.
- **Emergency Contact Numbers:** The site emergency contact numbers must be clearly posted and known to all personnel.